

Applied Cyber Defense & Threat Analysis

ADVANCED TOPICS IN PHISHING, MALWARE, AND DIGITAL FORENSICS

Cyber Defense in Action: Phishing, Threat Intel, Malware & Forensics Labs.

Tool: UBUNTU, KALI, VIRUSTOTAL, WHOIS LOOKUP, SPAMHAUS, URLHAUS, TALOS INTELLIGENCE, EMAIL HEADER ANALYSIS, URLSCAN, ALIENVAULT OTX, THREATFOX, HYBRID ANALYSIS, CYBERCHEF, STRINGS UTILITY, HASHING (SHA256), AVML, AND SCP.

Focus : Phishing detection, threat intelligence enrichment, malware analysis, and digital forensics — providing hands-on experience in identifying, analyzing, and responding to cyber threats.

Scenarios Investigated:

Day 21: Suspicious PayPal lookalike domain (secure-paypai.com) used in phishing.

Day 22: Fake “Banco do Bradesco Livelo” email urging users to redeem points.

Day 23: Phishing email containing malicious executable, IP, and domain IOCs.

Day 24: Static malware analysis of the EICAR test file.

Day 25: Acquisition and validation of a Linux memory dump using AVML.

Day 21 – Phishing Analysis: Lookalike Domain Investigation

Objective:

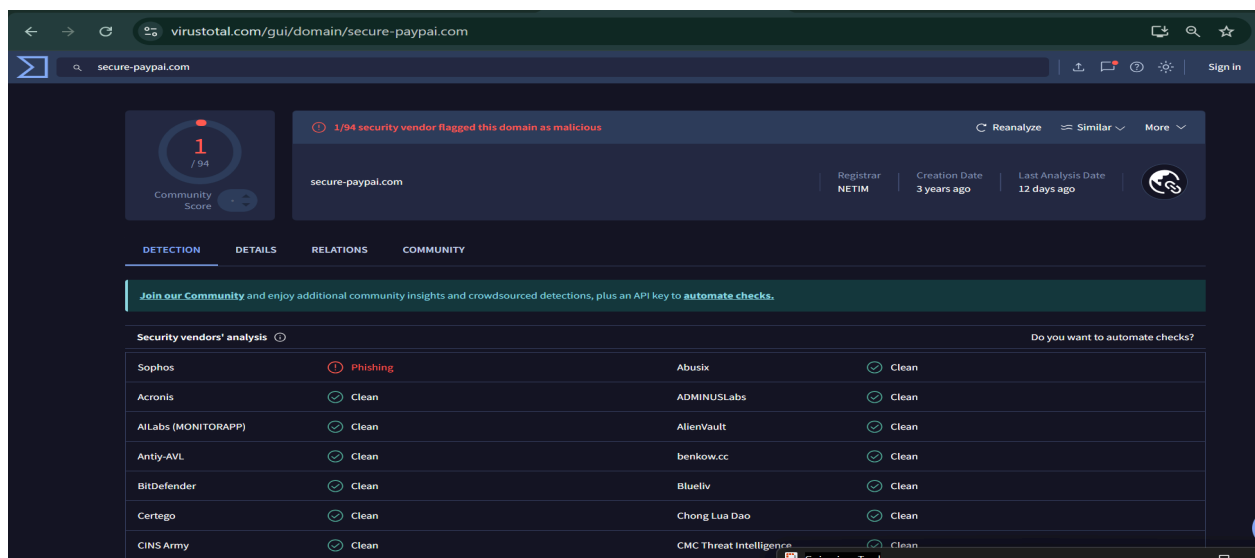
Analyze a suspicious PayPal phishing email and validate the sender's domain.

Task Performed:

Extracted domain secure-paypai.com from the email.

Checked domain reputation using VirusTotal, Whois Lookup, Spamhaus, URLhaus, and Talos Intelligence.

Verified SPF/DKIM/DMARC records and domain age.



Key Takeaways:

Recently registered domains with no authentication records are red flags.

Lookalike domains (typosquatting) are a common phishing tactic.

Reputation checks help confirm malicious intent.

Day 22 – Phishing Email Investigation: Fake Bank Alert

Objective:

Investigate a spoofed “Banco do Bradesco Livelo” points-expiration phishing email.

Task Performed:

Analyzed sender banco.bradesco@atendimento.com.br.

Extracted sender IP 137.184.34.4 and verified reputation.

Checked SPF authentication results.

Investigated embedded phishing link.

The screenshot shows the VirusTotal web interface for the IP address 137.184.34.4. The page displays a community score of 1/94, indicating a low reputation. A warning message states that 1/94 security vendors flagged this IP address as malicious. The IP is associated with AS 14061 (DIGITALOCEAN-ASN) and is located in the US. The last analysis date was 5 days ago. The page includes tabs for DETECTION, DETAILS, RELATIONS, and COMMUNITY. A table titled 'Security vendors' analysis' lists various security vendors and their results for this IP address.

Security vendors' analysis	Do you want to automate checks?
Criminal IP	Malicious
alphaMountain.ai	Suspicious
Abusix	Clean
Acronis	Clean
ADMINUSLabs	Clean
AILabs (MONITORAPP)	Clean
AlienVault	Clean
Antiy-AVL	Clean
benkow.cc	Clean
BitDefender	Clean
Blueliv	Clean
Certego	Clean
Chong Lua Dao	Clean
CINS Army	Clean

Security headers	
authentication-results	spf=temperor (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none;dmarc=temperor action=none header.from=atendimento.com.br;compauth=fail reason=001
X headers	
x-ms-exchange-crosstenant-originalarrivaltime	19 Sep 2023 18:36:44.1298 (UTC)
x-incomingheadercount	9
x-ms-exchange-crosstenant-authsource	BNSNAM11FT066.eop-nam11.prod.protection.outlook.com
x-ms-exchange-processed-by-bccfolding	15.20.6792.025
x-message-info	qZelhlYnHgoJocAkqKqrbZlellfpwpho8GvwdeJBPKcSpf16IG5llgDUPoij+JWsjp0BAMUgrOcbasZfN7V98T2UZUz54kBBX/DcD87QJ02aouiq4ENv4R1M3zDKP/XN09+50x9B66onJUDV4eqq36VUizqAa0zCzkWpdlEY9DzNE1OkawjHAizeUyMZ/UJK/Pz9dHAZA==
x-optenantaattributedmessage	84d9e7f1-e9f6-40af-b435-aaaaaaaaaaad
x-ms-exchange-organization-expiresint	1:00:00:00.0000000
x-sid-pra	BANCO.BRADESCO@ATENDIMENTO.COM.BR
x-microsoft-antispam	BCL9;
x-ms-exchange-organization-network-message-id	b9106deb-bd54-4815-e5c9-08dbb93f5fab
x-ms-exchange-organization-pcl	2
x-ms-exchange-sopdirect	true
x-ms-exchange-crosstenant-auth	Anonymous

[Pricing](#)
[Tools](#)
[Delivery Center](#)
[Monitoring](#)
[Products](#)
[Blog](#)
[Support](#)
[Login](#)

[SuperTool](#)
[MX Lookup](#)
[Blacklists](#)
[DMARC](#)
[Diagnostics](#)
[Email Health](#)
[DNS Lookup](#)
[Analyze Headers](#)
[All Tools](#)

Header Analyzed

Email Subject: CLIENTE PRIME - BRADESCO NIVEL: Seu cartão tem 92.990 pontos NIVEL expirando hoje!

[Analyze New Header](#)

Copy/Paste Warning
Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our [Email Deliverability tool](#)

Delivery Information

- DMARC Compliant (No DMARC Record Found)
 - SPF Alignment
 - SPF Authenticated
 - DKIM Alignment
 - DKIM Authenticated

Relay Information

Received 57 seconds

[mxtoolbox.com/Public/Tools/EmailHeaders.aspx?uid=36df9483-3749-407a-a78c-d9ef914685ce](#)

SPF and DKIM Information

dmarc:atendimento.com.br
[Show](#)
[Solve Email Delivery Problems](#)

Sorry, we couldn't find any name servers for '_dmarc.atendimento.com.br'

spf:ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06:::1
[Show](#)
[Solve Email Delivery Problems](#)

Sorry, we couldn't find any name servers for 'ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06'

DKIM Signature Error!
No DKIM-Signature header found - [more info](#)

DKIM Signature Error!
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - [more info](#)

Headers Found

Header Name	Header Value
Authentication-Results	spf=temperor (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none;dmarc=temperor action=none header.from=atendimento.com.br;compauth=fail reason=001

Key Takeaways:

SPF failures signal potential spoofing.

Professional-looking emails often crumble under header analysis.

Urgency and fear tactics remain common phishing strategies.

Day 23 – Threat Intelligence Basics: IOC Extraction

Objective:

Extract and enrich Indicators of Compromise (IOCs) from a phishing campaign.

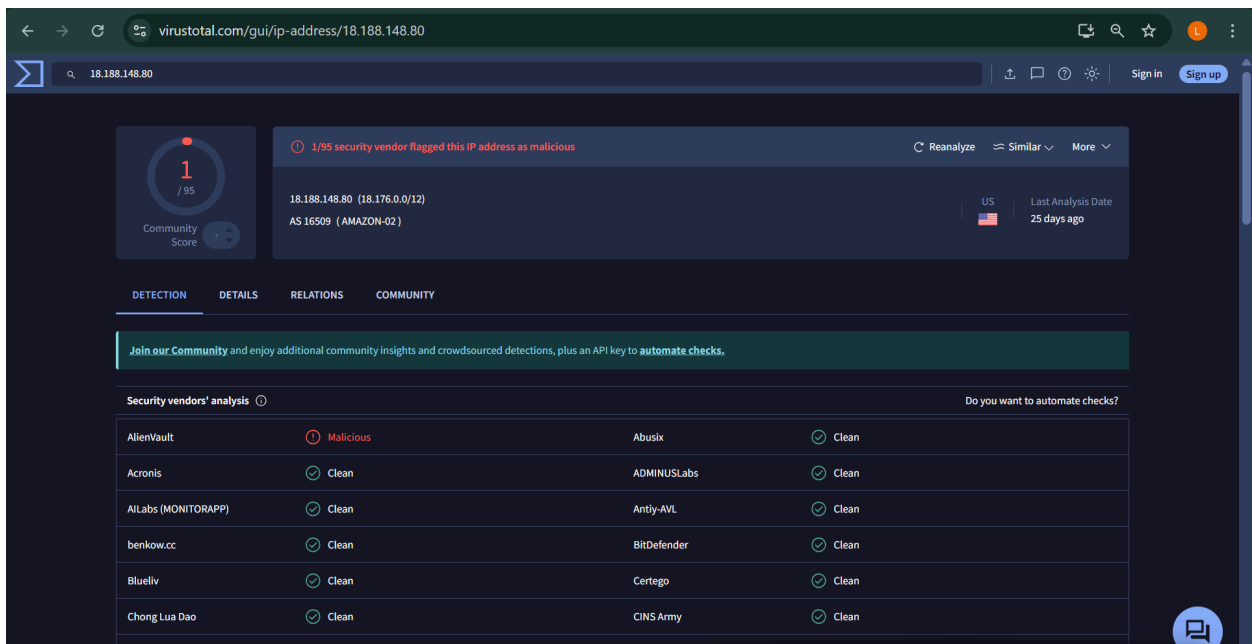
Task Performed:

Collected malicious file hash (SHA256), IP, and domain.

Investigated with VirusTotal, URLScan, AlienVault OTX, and ThreatFox.

Verified file as TROJAN.BADJOKE/ABOBUS.

Mapped infrastructure to AWS-hosted IP and suspicious domain.



The screenshot shows the VirusTotal web interface for the IP address 18.188.148.80. The browser address bar shows the URL `virustotal.com/gui/ip-address/18.188.148.80`. The page features a dark theme. At the top, a summary card displays a "Community Score" of 1/95, a warning that "1/95 security vendor flagged this IP address as malicious", and the IP details: "18.188.148.80 (18.176.0.0/12)" and "AS 16509 (AMAZON-02)". It also shows the location as "US" and the "Last Analysis Date" as "25 days ago". Below this, a navigation bar includes tabs for "DETECTION", "DETAILS", "RELATIONS", and "COMMUNITY". A green banner encourages joining the community. The main section, titled "Security vendors' analysis", contains a table of results from various vendors. A link to "Join our Community" is present. A table lists security vendors and their analysis results for the IP address 18.188.148.80. The table has two columns for vendor names and their corresponding status (Malicious or Clean).

Security vendors' analysis	
AlienVault	Malicious
Acronis	Clean
AllLabs (MONITORAPP)	Clean
benkow.cc	Clean
Blueliv	Clean
Chong Lua Dao	Clean
Abusix	Clean
ADMINUSLabs	Clean
Antiy-AVL	Clean
BitDefender	Clean
Certego	Clean
CINS Army	Clean

The screenshot shows the VirusTotal domain analysis page for `aaronthompson.ug`. The interface features a dark theme with a top navigation bar. On the left, a circular progress indicator shows a Community Score of 9/95. The main header area displays a warning icon and the text "9/95 security vendors flagged this domain as malicious". Below this, the domain name `aaronthompson.ug` is shown, along with a "Last Analysis Date" of 17 days ago and a "Reanalyze" button. The "DETECTION" tab is selected, showing a table of security vendors' analysis results. The table lists various vendors and their detection status for the domain.

Security vendors' analysis	Do you want to automate checks?
alphaMountain.ai	Malicious
BitDefender	Malware
CyRadar	Malicious
Lionic	Malicious
SOCRadar	Malware
Acronis	Clean
AllLabs (MONITORAPP)	Clean
Antiy-AVL	Clean
Blueliv	Clean
CheckUp	Clean
AlphaSOC	Malware
CRDF	Malicious
G-Data	Malware
Seclookup	Malicious
Abusix	Clean
ADMINUSLabs	Clean
AlienVault	Clean
benkow.cc	Clean
Certego	Clean
CMF	Clean

The screenshot shows the VirusTotal file analysis page for a file with SHA256 hash `d45a079c59c2860f9cf4578a8fc9f5fe8009cff8aaa83c572474d6bfe15ba95b`. The interface features a dark theme with a top navigation bar. On the left, a circular progress indicator shows a Community Score of 40/63. The main header area displays a warning icon and the text "40/63 security vendors flagged this file as malicious". Below this, the file name `flv.exe-source-code.Tz` is shown, along with a "Last Analysis Date" of 1 month ago and a "Reanalyze" button. The "DETECTION" tab is selected, showing a table of security vendors' analysis results. The table lists various vendors and their detection status for the file. The "BEHAVIOR" tab is also visible, showing a list of threat labels and categories.

Security vendors' analysis	Do you want to automate checks?
AhnLab-V3	Trojan.Win.Generic.C5753497
ALYac	Trojan.GenericKD.76183963
Avast	MBR:Abobus-A [Rtk]
Avira (no cloud)	TR/BadJoke.rjbli
CTX	Tzip.trojan.badjoke
AllCloud	Trojan:Win/BadJoke.QW
Arcabit	Trojan.Generic.D48A7997
AVG	MBR:Abobus-A [Rtk]
BitDefender	Trojan.GenericKD.76183959
Cynet	Malicious (score: 99)

Key Takeaways:

Threat Intelligence tools enrich raw IOCs for context.

Community platforms (OTX, ThreatFox) provide correlation with ongoing campaigns.

Malicious infrastructure often uses cloud hosting services.

Day 24 – Malware Analysis: Static Inspection

Objective:

Perform static analysis on a safe malware sample (EICAR test file).

Task Performed:

Generated SHA256 hash of the EICAR file.

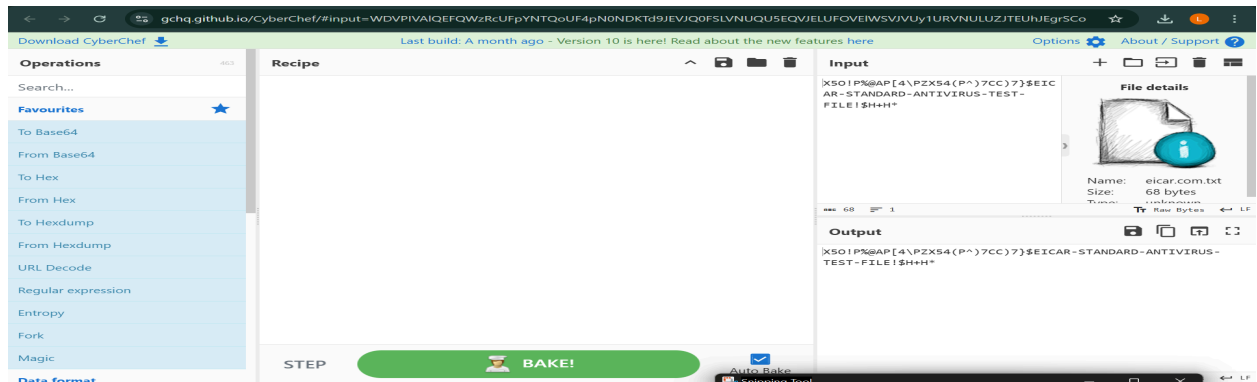
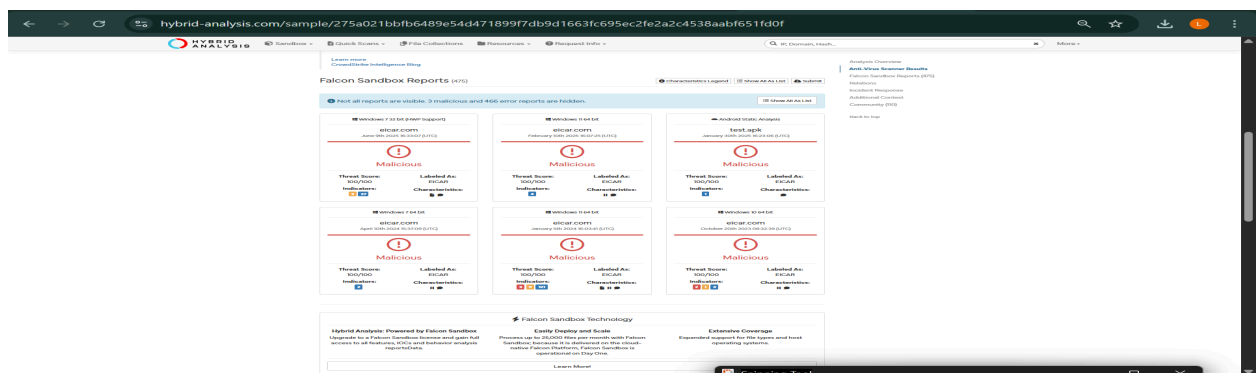
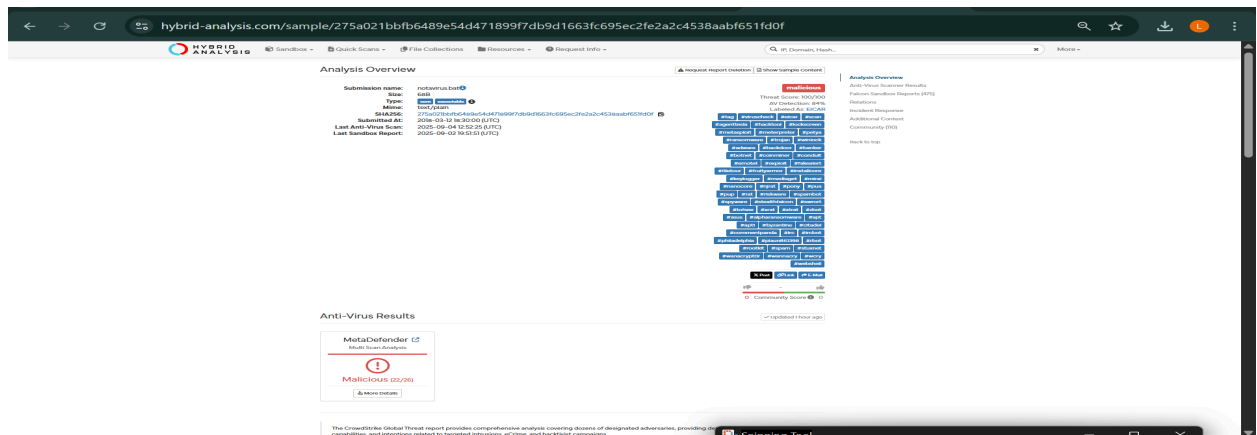
Verified detection across multiple AV engines on VirusTotal.

Used Strings Utility and CyberChef to inspect file contents.

Uploaded sample to Hybrid Analysis for sandbox simulation.

The screenshot shows the VirusTotal web interface for a file analysis. The file is identified as 'EICAR.txt' with a SHA256 hash of '275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f'. The file size is 68 B and it was last analyzed 8 minutes ago. The interface displays a 'Community Score' of 64/68 and a 'Reanalyze' button. Below the file information, there are tabs for 'DETECTION', 'DETAILS', 'RELATIONS', 'BEHAVIOR', and 'COMMUNITY'. The 'DETECTION' tab is active, showing a 'Join our Community' message and 'Code insights' about the EICAR test file. A table titled 'Security vendors' analysis' lists various antivirus engines and their detection results. The table has columns for the vendor, the detected threat name, and the family label. The vendors listed include AhnLab-V3, AliCloud, Antiy-AVL, Avast, AVG, Baidu, Alibaba, ALYac, Arcabit, Avast-Mobile, Avira (no cloud), and BitDefender. The detected threats are 'Virus/EICAR_Test_File', 'Engtest:Multi/Eicar', 'TestFile/Win32_EICAR', 'EICAR-Test-File (not A Virus)', 'Eicar', and 'Eicar-Test-Signature'. The family labels are 'Virus:Win32/EICAR.A', 'Misc:Eicar-Test-File', and 'Eicar-Test-Signature'.

Vendor	Threat Name	Family Label
AhnLab-V3	Virus/EICAR_Test_File	Virus:Win32/EICAR.A
AliCloud	Engtest:Multi/Eicar	Misc:Eicar-Test-File
Antiy-AVL	TestFile/Win32_EICAR	EICAR-Test-File (not A Virus)
Avast	EICAR-Test-NOT_Virus!!!	Eicar
AVG	EICAR-Test-NOT_Virus!!!	Eicar-Test-Signature
Baidu	Win32_Test_Eicar.a	



Key Takeaways:

Hashing and AV checks provide quick file reputation validation.

Strings extraction reveals useful embedded markers.

Sandboxing simulates behavior without real infection risks.

Day 25 – Digital Forensics: Memory Acquisition

Objective:

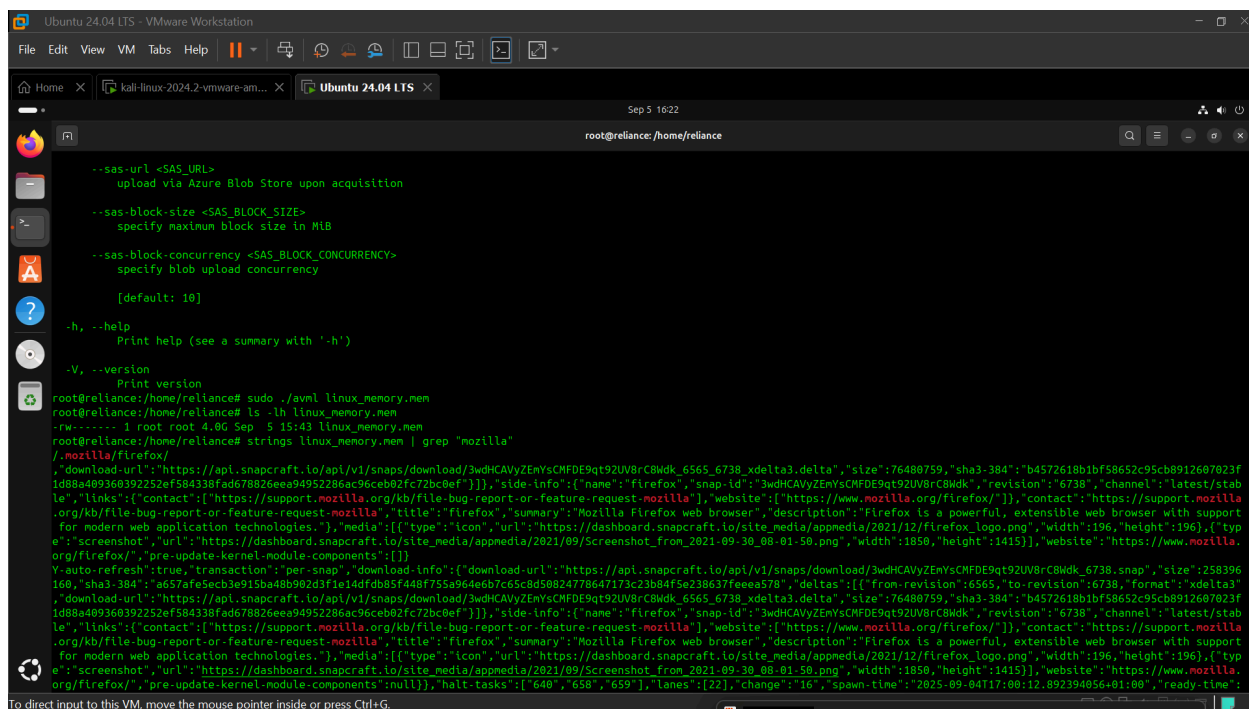
Capture and validate volatile memory from a Linux system for forensic analysis.

Task Performed:

Used AVML to acquire memory dump.

Validated dump with Strings Utility by searching for browser artifacts.

Transferred dump securely via SCP for further analysis.



```
Ubuntu 24.04 LTS - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2024.2-vmware-am... Ubuntu 24.04 LTS
Sep 5 16:22
root@reliance:/home/reliance

--sas-url <SAS_URL>
    upload via Azure Blob Store upon acquisition

--sas-block-size <SAS_BLOCK_SIZE>
    specify maximum block size in MiB

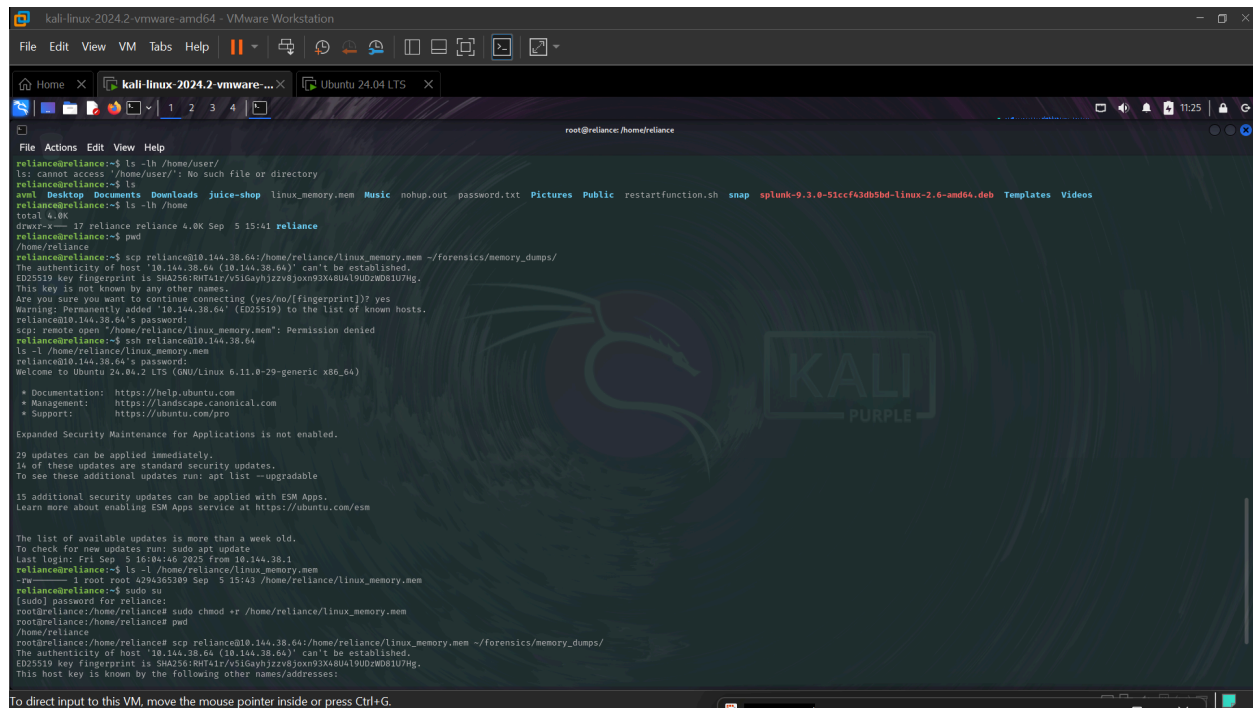
--sas-block-concurrency <SAS_BLOCK_CONCURRENCY>
    specify blob upload concurrency

    [default: 10]

-h, --help
    Print help (see a summary with '-h')

-V, --version
    Print version

root@reliance:/home/reliance# sudo ./avml linux_memory.mem
root@reliance:/home/reliance# ls -lh linux_memory.mem
-rw-r----- 1 root root 4.0G Sep  5 15:43 linux_memory.mem
root@reliance:/home/reliance# strings linux_memory.mem | grep "mozilla"
./mozilla/firefox/
, "download-url": "https://api.snapcraft.io/api/v1/snaps/download/3wdHCAVyZEnYsCMFDE9qt92UV8rC8Wdk_6565_6738_xdelta3.delta", "size": 76480759, "sha3-384": "b4572618b1bf58652c95cb8912607023f1d88a409360392252ef584338fad678826ee94952286ac96ceb02fc72bc0ef"}], "side-info": [{"name": "firefox", "snap-id": "3wdHCAVyZEnYsCMFDE9qt92UV8rC8Wdk", "revision": "6738", "channel": "latest/stable", "links": [{"contact": [{"https://support.mozilla.org/kb/file-bug-report-or-feature-request-mozilla"}, {"website": [{"https://www.mozilla.org/firefox/"}]}, {"contact": "https://support.mozilla.org/kb/file-bug-report-or-feature-request-mozilla", "title": "Firefox", "summary": "Mozilla Firefox web browser", "description": "Firefox is a powerful, extensible web browser with support for modern web application technologies."}], "media": [{"type": "icon", "url": "https://dashboard.snapcraft.io/site_media/appmedia/2021/12/firefox_logo.png", "width": 196, "height": 196}, {"type": "screenshot", "url": "https://dashboard.snapcraft.io/site_media/appmedia/2021/09/Screenshot_from_2021-09-30_08-01-50.png", "width": 1850, "height": 1415}], "website": "https://www.mozilla.org/firefox/"}], "pre-update-kernel-module-components": []}]
Y-auto-refresh": true, "transaction": "per-snap", "download-info": {"download-url": "https://api.snapcraft.io/api/v1/snaps/download/3wdHCAVyZEnYsCMFDE9qt92UV8rC8Wdk_6738.snap", "size": 258396160, "sha3-384": "ae57afe5ecb3e915ba48b902d3f1e14dfdb85f448f755a964e6b7c65cd8d50824778647173c23b84f5e238637f7eeea578", "deltas": [{"from-revision": 6565, "to-revision": 6738, "format": "xdelta3"}, {"download-url": "https://api.snapcraft.io/api/v1/snaps/download/3wdHCAVyZEnYsCMFDE9qt92UV8rC8Wdk_6565_6738_xdelta3.delta", "size": 76480759, "sha3-384": "b4572618b1bf58652c95cb8912607023f1d88a409360392252ef584338fad678826ee94952286ac96ceb02fc72bc0ef"}], "side-info": [{"name": "firefox", "snap-id": "3wdHCAVyZEnYsCMFDE9qt92UV8rC8Wdk", "revision": "6738", "channel": "latest/stable", "links": [{"contact": [{"https://support.mozilla.org/kb/file-bug-report-or-feature-request-mozilla"}, {"website": [{"https://www.mozilla.org/firefox/"}]}, {"contact": "https://support.mozilla.org/kb/file-bug-report-or-feature-request-mozilla", "title": "Firefox", "summary": "Mozilla Firefox web browser", "description": "Firefox is a powerful, extensible web browser with support for modern web application technologies."}], "media": [{"type": "icon", "url": "https://dashboard.snapcraft.io/site_media/appmedia/2021/12/firefox_logo.png", "width": 196, "height": 196}, {"type": "screenshot", "url": "https://dashboard.snapcraft.io/site_media/appmedia/2021/09/Screenshot_from_2021-09-30_08-01-50.png", "width": 1850, "height": 1415}], "website": "https://www.mozilla.org/firefox/"}], "pre-update-kernel-module-components": null}], "halt-tasks": [{"640", "650", "659"}], "lanes": [22], "change": 16, "spawn-time": 2025-09-04T17:00:12.892394056+01:00, "ready-time":
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```



The screenshot shows a Kali Linux virtual machine window titled "kali-linux-2024.2-vmware-amd64 - VMware Workstation". The terminal window displays the following commands and output:

```
reliance@reliance:~$ ls -lh /home/user/
ls: cannot access '/home/user/': No such file or directory
reliance@reliance:~$ ls
.  ..  Desktop  Documents  Downloads  juice-shop  linux_memory.mem  Music  nohup.out  password.txt  Pictures  Public  restartfunction.sh  snap  splunk-9.3.0-51ccfa3db5bd-linux-2.6-amd64.deb  Templates  Videos
reliance@reliance:~$ ls -lh /home
total 4.0K
drwxr-xr-x 17 reliance reliance 4.0K Sep  5 15:41 reliance
reliance@reliance:~$ pwd
/home/reliance
reliance@reliance:~$ scp reliance@10.144.38.64:/home/reliance/linux_memory.mem ~/forensics/memory_dumps/
The authenticity of host '10.144.38.64 (10.144.38.64)' can't be established.
ED25519 key fingerprint is SHA256:8u7f4tz/65iGaj3jzve83omSP94u41sU0zw081U7Hg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.144.38.64' (ED25519) to the list of known hosts.
reliance@10.144.38.64's password:
scp: remote open "/home/reliance/linux_memory.mem": Permission denied
reliance@reliance:~$ ssh reliance@10.144.38.64
ls -l /home/reliance/linux_memory.mem
reliance@10.144.38.64's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-29-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
14 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

15 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Fri Sep  5 16:04:46 2025 from 10.144.38.1
reliance@reliance:~$ ls -l /home/reliance/linux_memory.mem
-rw-r--r-- 1 root root 4294365389 Sep  5 15:43 /home/reliance/linux_memory.mem
reliance@reliance:~$ sudo su
[sudo] password for reliance:
root@reliance:/home/reliance# chmod +r /home/reliance/linux_memory.mem
root@reliance:/home/reliance# pwd
/home/reliance
root@reliance:/home/reliance# scp reliance@10.144.38.64:/home/reliance/linux_memory.mem ~/forensics/memory_dumps/
The authenticity of host '10.144.38.64 (10.144.38.64)' can't be established.
ED25519 key fingerprint is SHA256:8u7f4tz/65iGaj3jzve83omSP94u41sU0zw081U7Hg.
This host key is known by the following other names/addresses:
```

Key Takeaways:

Memory acquisition preserves volatile evidence (processes, keys, network sessions).

Validation ensures dump integrity before analysis.

Secure transfer methods prevent evidence tampering.

SUMMARY

Over the course of Days 21 to 25, I gained practical, hands-on experience across the core areas of phishing analysis, threat intelligence, malware inspection, and digital forensics.

I began with the investigation of phishing emails and lookalike domains, where I learned how to identify red flags such as recently registered domains, failed SPF/DKIM/DMARC checks, and urgency-driven messages (Days 21–22). I then advanced into threat intelligence enrichment, extracting IOCs from phishing campaigns and validating them using community-driven tools such as VirusTotal, OTX, and ThreatFox (Day 23).

Building on this, I explored the fundamentals of malware analysis through static inspection of the EICAR test file, applying techniques like hashing, strings extraction, and sandbox simulation (Day 24). Finally, I completed a digital forensics exercise, acquiring a Linux memory dump with AVML, validating its integrity, and securely transferring it for further analysis (Day 25).

This phase of my SOC Analyst Lab Journey provided me with a holistic understanding of how to detect, analyze, and respond to cyber threats. It reinforced the importance of combining phishing awareness, threat intelligence, malware analysis, and forensic techniques to strengthen organizational security and incident response capabilities.